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Water outlet device

Abstract

The present invention discloses a water outlet device, comprising a fluid diverting mechanism, a seal having elasticity, a water inlet mechanism having a water inlet member with elasticity at a water inlet end, and a diversion mechanism rotatably connected to the water inlet mechanism; the water inlet mechanism comprises a water inlet channel, and the fluid diverting mechanism comprises a first water outlet channel; the diversion mechanism comprises a second water outlet channel; the seal is capable of placing the water inlet channel in communication with the first water outlet channel or the second water outlet channel when the diversion mechanism is movable relative to the fluid diverting mechanism. The present invention improves the yield of the product by absorbing the tolerances of the water outlet device as a whole by means of the seal with elasticity and the deformation of the water inlet member with elasticity.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2014/0217202	12/2013	Stein	239/392	E03C 1/08
2014/0217314	12/2013	Kao	251/12	F16K 21/06
2019/0194918	12/2018	Schurle	N/A	E03C 1/084
2020/0263806	12/2019	Kao	N/A	E03C 1/08
2021/0148100	12/2020	Zhou	N/A	E03C 1/0405
2022/0213977	12/2021	Kao	N/A	E03C 1/0404

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103120990	12/2012	CN	B05B 1/16
111804448	12/2019	CN	N/A
111827417	12/2019	CN	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the benefit of Chinese Patent Application Nos. 202220205269.3 and 202210087859.5 filed on Jan. 25, 2022. All the above are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to the field of bathroom and in particular to a water outlet device.

BACKGROUND ART

(3) At present, there are a variety of water outlet devices on the market, which are installed at the

water outlet of sanitary products to control or regulate the water flow amount and state.

(4) However, the existing water outlet devices have the defects of single function, complicated structure and being easy to be damaged by frequent state conversion.

(5) In view of the above-mentioned defects, Chinese patent documents disclose a multi-functional water outlet device (Application No. 202010646138.4) and a water outlet device (Application No. 202010646694.1). Since the two water outlet devices have a fluid diverting mechanism owing more matching points with other parts in the axial direction, there is a problem of excessive accumulated tolerance in the assembly, so that a large amount of interference occurs between the fluid diverting mechanism and the water inlet component during the matching, resulting in that the water passage cannot be completely switched into place, thus resulting in that the water outlet function cannot be completely switched and the product yield is low.

SUMMARY OF THE INVENTION

(6) The technical problem to be solved by the present invention is as follows. A water outlet device is provided to reduce product tolerances and improve product yield.

(7) In order to solve the above technical problem, the technical solution adopted by the present invention is as follows: a water outlet device comprises a fluid diverting mechanism, a seal having elasticity, a water inlet mechanism having a water inlet member with elasticity at a water inlet end, and a diversion mechanism rotatably connected to the water inlet mechanism; the water inlet mechanism comprises a water inlet channel; the fluid diverting mechanism comprises a first water outlet channel; the diversion mechanism comprises a second water outlet channel; the seal is mounted to a water inlet end of the fluid diverting mechanism, and the seal is capable of placing the water inlet channel in communication with the first water outlet channel or the second water outlet channel when the diversion mechanism is movable relative to the fluid diverting mechanism; and when the water inlet channel communicates with the second water outlet channel, the seal can be pressed against a side of the water inlet member.

(8) The advantageous effect of the present invention is that a seal having elasticity is provided at the water inlet end of the fluid diverting mechanism, and a water inlet member having elasticity is provided at the water inlet end of the water inlet mechanism, so that the seal can flexibly contact with the water inlet member when the seal performs relative movement with the fluid diverting mechanism, thereby enabling the tolerance generated during assembly of the water outlet device to be absorbed by deformation between the seal and the water inlet member. Thereby, the invention ensures that it is switched in place each time the communication relationship between the water inlet channel and the first water outlet channel and the second water outlet channel, so as to achieve the purpose of improving the product yield.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view showing the structure of a water outlet device according to the present invention;

(2) FIG. 2 is an exploded view of the water outlet device of the present invention;

(3) FIG. 3 is an exploded view of a water outlet device according to the present invention;

(4) FIG. 4 is a cross-sectional view showing a state where a water inlet channel of the water outlet device of the present invention communicates with a first water outlet channel;

(5) FIG. 5 is an enlarged view of a portion A of FIG. 4;

(6) FIG. 6 is a sectional view showing a state where the water inlet channel of the water outlet device of the present invention communicates with a second water outlet channel;

(7) FIG. 7 is a schematic view showing the structure of a diversion mechanism according to the present invention.

DESCRIPTION OF REFERENCE NUMERALS

(8) **1**, water inlet mechanism; **10**, water inlet channel; **11**, water diverting body; **12**, water outlet body; **13**, water inlet member; **131**, water inlet net; **132**, water outlet hole; **2**, diversion mechanism; **20**, second water outlet channel; **21**, mounting hole; **22**, step; **221**, first step surface; **222**, second step surface; **3**, seal; **30**, buffer space **31**, first diversion slope; **32**, second diversion slope; **4**, fluid diverting mechanism; **40**, first water outlet channel; **41**, fluid diverting body; **411**, inlet; **412**, protrusion; **413**, first limiting stage; **414**, second limiting stage; **42**, reset member; **43**, functional member; **5**, elastic seal; **6**, housing; **7**, water outlet rectifying net.

DETAILED DESCRIPTION OF THE INVENTION

(9) In order to explain the technical contents, achieved objects and effects of the present invention in detail, a description is made in combination with the embodiments and the accompanying drawings.

(10) With reference to FIGS. **1-7**, a water outlet device includes a fluid diverting mechanism **4**, a seal **3** having elasticity, a water inlet mechanism **1** having a water inlet member **13** with elasticity at a water inlet end, and a diversion mechanism **2** rotatably connected to the water inlet mechanism **1**.

(11) The water inlet mechanism **1** includes a water inlet channel **10**.

(12) The fluid diverting mechanism **4** includes a first water outlet channel **40**.

(13) The diversion mechanism **2** includes a second water outlet channel **20**.

(14) The seal **3** is mounted to a water inlet end of the fluid diverting mechanism **4**, and the seal **3** is capable of placing the water inlet channel **10** in communication with the first water outlet channel **40** or the second water outlet channel **20** when the diversion mechanism **2** is movable relative to the fluid diverting mechanism **4**.

(15) When the water inlet channel **10** communicates with the second water outlet channel **20**, the seal **3** can be pressed against a side of the water inlet member **13**.

(16) The working principle of the present invention is as follows.

(17) By providing the water inlet member **13** having elasticity and the seal **3** having elasticity, when the diversion mechanism **2** moves relative to the fluid diverting mechanism **4**, the water inlet member **13** and the seal **3** flexibly contact to generate deformation so as to absorb the tolerance of the water outlet device as a whole, thereby improving the reliability of the water outlet channel switching and improving the product yield.

(18) As known from the above description, the present invention has following advantageous effects. A seal **3** is provided at the water inlet end of the fluid diverting mechanism **4**, and a water inlet member **13** is provided at the water inlet end of the water inlet mechanism **1**, so that the seal **3** can flexibly contact with the water inlet mechanism **1** when the seal **3** performs relative movement with the fluid diverting mechanism **4**, or the water inlet member **13** can contact with the fluid diverting mechanism **4** when the fluid diverting mechanism **4** performs relative movement with the diversion mechanism **2**, thereby enabling it to absorb the tolerance generated via deformation between the seal **3** and the water inlet member **13** or the fluid diverting mechanism **4** and the water inlet member **13** when the water outlet device is assembled. Furthermore, the invention ensures that it is switched in place each time the communication relationship between the water inlet channel **10** and the first water outlet channel **40** and the second water outlet channel **20**, so as to achieve the purpose of improving the product yield.

(19) Further, an elastic seal **5** is embedded in a water outlet end of the water inlet mechanism **1**.

(20) When the water inlet channel **10** communicates with the first water outlet channel **40**, the fluid diverting mechanism **4** can be pressed against the elastic seal **5**.

(21) It can be seen from the above-mentioned description that the elastic seal **5** is provided, on the one hand, for providing an axial supporting force for the fluid diverting mechanism **4** when the water inlet channel **10** communicates with the first water outlet channel **40**. The elastic seal **5** can also generate deformation for absorbing tolerances. On the other hand, the elastic seal **5** is provided for improving the sealing performance between the fluid diverting mechanism **4** and the water inlet

mechanism **1** and reducing the probability of water leakage.

(22) Further, the water inlet member **13** has a water inlet net **131** recessed in an axial direction along a water inlet end of the water inlet mechanism **1** to a water outlet end of the water inlet mechanism **1**.

(23) A plurality of water outlet holes **132** are provided around the side wall in the radial direction of the water inlet net **131**.

(24) It can be seen from the above-mentioned description that a plurality of water outlet holes **132** are provided around the side wall of the water inlet net **131**, so that the water flows into the water inlet mechanism **1** from the side surface of the water inlet net **131** to perform rectification and filtration.

(25) Further, the side wall in the radial direction of the water inlet net **131** inclines gradually and concavely along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**.

(26) As can be seen from the above description, the side wall in the radial direction of the water inlet net **131** is inclined to provide a larger deformation space for the water inlet member **13** to absorb larger tolerances.

(27) Further, the fluid diverting mechanism **4** includes a fluid diverting body **41**.

(28) The seal **3** is sleeved outside a water inlet end of the fluid diverting body **41**.

(29) Further, a first limiting stage **413** is provided outside the water inlet end of the fluid diverting body **41**, and the seal **3** is sleeved outside the first limiting stage **413**.

(30) Further, the water inlet end of the fluid diverting body **41** is further provided with a second limiting stage **414**. The first limiting stage **413** and the second limiting stage **414** are provided in sequence along the axial direction of the fluid diverting body **41**. The seal **3** can be pressed against the second limiting stage **414**.

(31) It can be seen from the above-mentioned description that the first limiting stage **413** and the second limiting stage **414** are provided for improving the tightness of the connection between the seal and the fluid diverting mechanism **4**.

(32) Further, a side of the seal **3** facing towards the water inlet end of the water inlet mechanism **1** has a first diversion slope **31**. A side of the seal **3** facing towards the water outlet end of the water inlet mechanism **1** has a second diversion slope **32**.

(33) The first diversion slope **31** inclines gradually and convexly along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**.

(34) The second diversion slope **32** inclines gradually and convexly along the water outlet end of the water inlet mechanism **1** to the water inlet end of the water inlet mechanism **1**.

(35) When the water inlet channel **10** communicates with the first water outlet channel **40**, the second diversion slope **32** is pressed against the water inlet mechanism **1**.

(36) It can be seen from the above-mentioned description that the first diversion slope **31** and the second diversion slope **32** are provided for reducing the resistance received by the seal **3** in the axial direction when the fluid diverting mechanism **4** moves in the axial direction, and improving the fluency when switching in different water outlet states.

(37) Further, the top of the seal **3** and the top of the fluid diverting body **41** form a buffer space **30** in the axial direction of the fluid diverting body **41**.

(38) As can be seen from the above description, the buffer space **30** is formed for providing a larger deformation space for the seal **3** to absorb larger tolerances.

(39) Further, the fluid diverting mechanism **4** includes a fluid diverting body **41**, a reset member **42**, and a functional member **43**.

(40) The reset member **42** and the functional member **43** are successively arranged in the fluid diverting body **41** along the water inlet end of the fluid diverting body **41** to the water outlet end of the fluid diverting body **41**.

(41) Both ends of the reset member **42** are pressed against the water inlet end of the water inlet

mechanism **1** and the functional member **43**, respectively.

(42) It can be seen from the above-mentioned description that the reset member **42** is provided for providing a reset driving force in an axial direction for the fluid diverting body **41** to ensure reliability when switching in different water outlet states. The functional member **43** is provided for enabling the water flowing out from the first water outlet channel **40** to achieve different water outlet states so as to meet different requirements of a user.

Embodiment 1

(43) With reference to FIGS. 1-7, a water outlet device includes a water inlet mechanism **1** and a diversion mechanism **2** rotatably connected to the water inlet mechanism **1**. The water inlet mechanism **1** includes a water inlet channel **10**, and further includes a fluid diverting mechanism **4** and a seal **3** with elasticity. The fluid diverting mechanism **4** is installed in the water inlet mechanism **1** and is arranged through the axial center of the diversion mechanism **2**. The fluid diverting mechanism **4** includes a first water outlet channel **40**. The diversion mechanism **2** includes a second water outlet channel **20**. During the movement of the diversion mechanism **2** relative to the fluid diverting mechanism **4**, the water inlet channel **10** is brought into communication with either the first water outlet channel **40** or the second water outlet channel **20** via the seal **3**.

(44) With reference to FIGS. 2-5, an elastic seal **5** is embedded in the water outlet end of the water inlet mechanism **1**. When the water inlet channel **10** communicates with the first water outlet channel **40**, the fluid diverting mechanism **4** can be pressed against the elastic seal **5**.

(45) With reference to FIGS. 2 and 3, the water inlet mechanism **1** includes a water diverting body **11**, a water outlet body **12** and a water inlet member **13**. The water inlet member **13** is embedded on the top of the fluid diverting body **41**. The fluid diverting mechanism **4** is successively arranged through the axial center of the fluid diverting body **41** and the water outlet body **12**. The fluid diverting body **41** is embedded in the water outlet body **12**.

(46) Referring to FIG. 2, the fluid diverting mechanism **4** includes a fluid diverting body **41**, a reset member **42**, and a functional member **43**. The reset member **42** and the functional part **43** are successively arranged in the fluid diverting body **41** along the water inlet end of the fluid diverting body **41** to the water outlet end of the fluid diverting body **41**. Both ends of the reset member **42** are respectively pressed against the water inlet end of the water inlet mechanism **1** and the functional member **43**. The seal **3** is sleeved outside the water inlet end of the fluid diverting body **41**. Preferably, the reset member **42** is a spring. The functional member **43** can be selected according to actual requirements.

(47) With reference to FIGS. 4 and 6, a side of the seal **3** facing towards the water inlet end of the water inlet mechanism **1** has a first diversion slope **31**. A side of the seal **3** facing towards the water outlet end of the water inlet mechanism **1** has a second diversion slope **32**. The first guide slope inclines gradually and convexly along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**. The second guide slope inclines gradually and convexly along the water outlet end of the water inlet mechanism **1** to the water inlet end of the water inlet mechanism **1**. When the water inlet channel **10** communicates with the first water outlet channel **40**, the second diversion slope **32** is pressed against an inlet **411** of the fluid diverting body **41**.

(48) Referring to FIG. 5, the top of the seal **3** and the top of the fluid diverting body **41** form a buffer space **30** in the axial direction of the fluid diverting body **41**. Preferably, the buffer space **30** has a height h of 0.3 mm to 0.7 mm in the axial direction. In the present embodiment, the height h of the buffer space **30** is preferably 0.5 mm.

(49) With reference to FIGS. 4 and 7, a mounting hole **21** for enabling the fluid diverting mechanism **4** to pass through is provided in the diversion mechanism **2**. A step **22** is provided on a side of the mounting hole **21** facing towards the water inlet mechanism **1**. Specifically, the step **22** includes a first step surface **221** and a second step surface **222**. The first step surface **221** and the

second step surface **222** are connected in a smooth transition. Two symmetrically provided protrusions **412** are provided on the outer side of the fluid diverting body **41**. The protrusions **412** can move along the first step surface **221** and the second step surface **222** so as to drive the fluid diverting body **41** to move in an axial direction. Furthermore, the communication relationship between the water inlet channel **10** and the first water outlet channel **40** and the second water outlet channel **20** is switched.

(50) Specific implementations of this embodiment are as follows.

(51) When the water inlet channel **10** is in communication with the first water outlet channel **40**, the seal **3** is pressed against an inlet **411** of the fluid diverting body **41**, and the second water outlet channel **20** is sealed. At this time, water flows through the water inlet member **13** and the fluid diverting body **41** successively, and flows out in functional water, such as spray water, tooth flushing water, blade water or transparent water, etc.

(52) By rotating the diversion mechanism **2**, the protrusion **412** of the fluid diverting body **41** moves along the step **22** so as to realize the switching of the communication relationship. When the water inlet channel **10** communicates with the second water outlet channel **20**, the seal **3** is pressed against a lower side of the water inlet member **13**. At this time, the water flow passes through the water inlet member **13**, the fluid diverting body **41**, the water outlet body **12** and the diversion mechanism **2** in sequence and is discharged. The sparkling water flows out at this time. The principle of flowing out the sparkling water is prior art, and will not be described in detail here.

Embodiment II

(53) With reference to FIGS. **1-7**, a water outlet device includes a fluid diverting mechanism **4**, and further includes a water inlet mechanism **1** having a water inlet member **13** with elasticity at a water inlet end, and a diversion mechanism **2** rotatably connected to the water inlet mechanism **1**. The water inlet mechanism **1** includes a water inlet channel **10**, and the fluid diverting mechanism **4** includes a first water outlet channel **40**. The diversion mechanism **2** includes a second water outlet channel **20**. When the diversion mechanism **2** moves relative to the fluid diverting mechanism **4**, the water inlet channel **10** is in communication with the first water outlet channel **40** or the second water outlet channel **20**. When the water inlet channel **10** communicates with the second water outlet channel **20**, the fluid diverting mechanism **4** can be pressed against a side of the water inlet member **13**.

(54) Referring to FIG. **4**, the water inlet member **13** has a water inlet net **131** recessed in the axial direction along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**. A plurality of water outlet holes **132** are provided around the side wall in the radial direction of the water inlet net **131**.

(55) Referring to FIG. **4**, the side wall in the radial direction of the water inlet net **131** inclines gradually and concavely along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**.

(56) Referring to FIG. **4**, a side of the water inlet net **131** facing towards the fluid diverting mechanism **4** is a plane. In the present embodiment, the water inlet member **13** is provided instead of the conventional filter assembly to achieve tolerance absorption under the premise of meeting the filtering requirements of the water outlet device, and to reduce the number of components in the water outlet device, thus reducing the production cost and improving the product yield.

(57) With reference to FIGS. **2** and **3**, the water inlet mechanism **1** includes a water diverting body **11** and a water outlet body **12**. The water inlet member **13** is embedded on the top of the water diverting body **11**. The fluid diverting mechanism **4** is successively arranged through the axial center of the water diverting body **11** and the water outlet body **12**. The water diverting body **11** is embedded in the water outlet body **12**.

(58) Referring to FIG. **2**, the fluid diverting mechanism **4** includes a fluid diverting body **41**, a reset member **42**, and a functional member **43**. The reset member **42** and the functional part **43** are successively arranged in the fluid diverting body **41** along the water inlet end of the fluid diverting

body **41** to the water outlet end of the fluid diverting body **41**. Both ends of the reset member **42** are respectively pressed against the water inlet end of the water inlet mechanism **1** and the functional member **43**. The seal is sleeved outside the water inlet end of the fluid diverting body **41**. Preferably, the reset member **42** is a spring. The functional member **43** can be selected according to actual requirements.

(59) Referring to FIG. 5, the top of the seal **3** and the top of the fluid diverting body **41** form a buffer space **30** in the axial direction of the fluid diverting body **41**. Preferably, the buffer space **30** has a height h of 0.3 mm to 0.7 mm in the axial direction. In the present embodiment, the height of the buffer space **30** is preferably 0.5 mm.

(60) With reference to FIGS. 4 and 7, a mounting hole **21** for enabling the fluid diverting mechanism **4** to pass through is provided in the diversion mechanism **2**. A step **22** is provided on a side of the mounting hole **21** facing towards the water inlet mechanism **1**. Specifically, the step **22** includes a first step surface **221** and a second step surface **222**. The first step surface **221** and the second step surface **222** are connected in a smooth transition. Two symmetrically provided protrusions **412** are provided on the outer side of the fluid diverting body **41**. The protrusions **412** can move along the first step surface **221** and the second step surface **222** so as to drive the fluid diverting body **41** to move in an axial direction. Furthermore, the communication relationship between the water inlet channel **10** and the first water outlet channel **40** and the second water outlet channel **20** is switched.

(61) The working procedure of this embodiment is as follows.

(62) When the water inlet channel **10** is in communication with the first water outlet channel **40**, the seal **3** is pressed against an inlet **411** of the fluid diverting body **41**, and the second water outlet channel **20** is sealed. At this time, water flows through the water inlet member **13** and the fluid diverting body **41** successively, and flows out in functional water, such as spray water, tooth flushing water, blade water or transparent water, etc.

(63) By rotating the diversion mechanism **2**, the protrusion **412** of the fluid diverting body **41** moves along the step **22** so as to realize the switching of the communication relationship. When the water inlet channel **10** communicates with the second water outlet channel **20**, the seal **3** is pressed against a lower side of the water inlet member **13**. At this time, the water flow passes through the water inlet member **13**, the fluid diverting body **41**, the water outlet body **12** and the diversion mechanism **2** in sequence and is discharged. The sparkling water flows out at this time. The principle of flowing out the sparkling water is prior art, and will not be described in detail here.

Embodiment III

(64) With reference to FIGS. 1-7, a water outlet device includes a fluid diverting mechanism **4**, a seal **3** having elasticity, a water inlet mechanism **1** having a water inlet member **13** at a water inlet end, and a diversion mechanism **2** rotatably connected to the water inlet mechanism **1**. The water inlet mechanism **1** includes a water inlet channel **10**, and the fluid diverting mechanism **4** includes a first water outlet channel **40**. The diversion mechanism **2** includes a second water outlet channel **20**. The seal **3** is mounted to a water inlet end of the fluid diverting mechanism **4**. When the diversion mechanism **2** moves relative to the fluid diverting mechanism **4**, the seal **3** can make the water inlet channel **10** communicate with the first water outlet channel **40** or the second water outlet channel **20**. When the water inlet channel **10** communicates with the second water outlet channel **20**, the seal **3** can be pressed against a side of the water inlet member **13**.

(65) With reference to FIGS. 2 and 6, an elastic seal **5** is embedded in the water outlet end of the water inlet mechanism **1**. When the water inlet channel **10** communicates with the first water outlet channel **40**, the fluid diverting mechanism **4** can be pressed against the elastic seal **5**.

(66) With reference to FIG. 2, the water inlet mechanism **1** includes a water diverting body **11**, a water outlet body **12** and a water inlet member **13**. The water inlet member **13** is embedded on the top of the water diverting body **11**. The fluid diverting mechanism **4** is successively arranged through the axial center of the fluid diverting body **41** and the water outlet body **12**. The water

diverting body **11** is embedded in the outlet body **12**.

(67) Referring to FIGS. **4** to **6**, the water inlet member **13** has a water inlet net **131** recessed in the axial direction along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**. A plurality of water outlet holes **132** are provided around the side wall in the radial direction of the water inlet net **131**. The water outlet holes **132** are strip-shaped, and the water outlet holes **132** extend from the water inlet end in the direction of the water outlet end. At the same time, the side wall in the radial direction of the water inlet net **131** gradually inclines towards the axial direction along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**. The bottom of the water inlet net **131** is planar, so as to improve the sealing performance between the water inlet net **131** and the seal **3** when the water inlet net **131** abuts against the seal **3** via the planar surface of the bottom, which prevents the water flow from leaking out from the first water outlet channel **40** when the second water outlet channel **20** discharges water. In the present embodiment, the water inlet member **13** is provided instead of the conventional filter assembly to better achieve tolerance absorption due to the strip-shaped water outlet hole **132** also capable of bringing more axial elastic deformation to the water inlet member **13** under the premise of meeting the filtering requirements of the water outlet device, and to reduce the number of components in the water outlet device, thus reducing the production cost and improving the product yield.

(68) Referring to FIG. **2**, the fluid diverting mechanism **4** includes a fluid diverting body **41**, a reset member **42**, and a functional member **43**. The reset member **42** and the functional part **43** are successively arranged in the fluid diverting body **41** along the water inlet end of the fluid diverting body **41** to the water outlet end of the fluid diverting body **41**. Both ends of the reset member **42** are respectively pressed against the water inlet end of the water inlet mechanism **1** and the functional member **43**. The seal **3** is sleeved outside the water inlet end of the fluid diverting body **41**.

(69) Referring to FIG. **6**, a first limiting stage **413** is provided outside the water inlet end of the fluid diverting body **41**, and the seal **3** is sleeved outside the first limiting stage **413**.

(70) Referring to FIG. **6**, the water inlet end of the fluid diverting body **41** is further provided with a second limiting stage **414**. The first limiting stage **413** and the second limiting stage **414** are provided in sequence along the axial direction of the fluid diverting body **41**. The seal **3** can be pressed against the second limiting stage **414**.

(71) With reference to FIG. **6**, a side of the seal **3** facing towards the water inlet end of the water inlet mechanism **1** has a first diversion slope **31**. A side of the seal **3** facing towards the water outlet end of the water inlet mechanism **1** has a second diversion slope **32**. The first diversion slope **31** inclines gradually and convexly along the water inlet end of the water inlet mechanism **1** to the water outlet end of the water inlet mechanism **1**. The second diversion slope **32** inclines gradually and convexly along the water outlet end of the water inlet mechanism **1** to the water inlet end of the water inlet mechanism **1**. When the water inlet channel **10** communicates with the first water outlet channel **40**, the second diversion slope **32** is pressed against an inlet **411** of the fluid diverting body **41**.

(72) Referring to FIG. **5**, the top of the seal **3** and the top of the fluid diverting body **41** form a buffer space **30** in the axial direction of the fluid diverting body **41**. Alternatively, the buffer space **30** in the axial direction has a height h of 0.3 mm to 1 mm, preferably 0.5 mm. The height h of the buffer space in the axial direction may be up to 1 mm in order to avoid the possibility of negative installation tolerances leading to an inability to completely close the first water outlet channel.

(73) With reference to FIGS. **4** and **7**, a mounting hole **21** for enabling the fluid diverting mechanism **4** to pass through is provided in the diversion mechanism **2**. A step **22** is provided on a side of the mounting hole **21** facing towards the water inlet mechanism **1**. Specifically, the step **22** includes a first step surface **221** and a second step surface **222**. The first step surface **221** and the second step surface **222** are connected in a smooth transition. Two symmetrically provided

protrusions **412** are provided on the outer side of the fluid diverting body **41**. The protrusions **412** can move along the first step surface **221** and the second step surface **222** so as to drive the fluid diverting body **41** to move in an axial direction. Furthermore, the communication relationship between the water inlet channel **10** and the first water outlet channel **40** and the second water outlet channel **20** is switched.

(74) With reference to FIGS. **1** and **6**, it further includes a housing **6** and a water outlet rectifying net **7**. The housing is sleeved outside the water inlet mechanism **1** and the diversion mechanism **2**. The water outlet rectifying net **7** is embedded in the water outlet end of the housing **6** and is clamped with the diversion mechanism **2**. A gap is left between the housing **6** and the water inlet mechanism **1**. The housing **6** is connected to the diversion mechanism **2** so as to drive the diversion mechanism **2** to rotate relative to the fluid diverting mechanism **4** via the housing **6**.

(75) In view of the above, the present invention provides a water outlet device, which is provided with a seal having elasticity, instead of the hard seal at the top of the fluid diverting mechanism in the prior art. On the premise of achieving normal switching of the water outlet channel, the tolerance is absorbed by the deformation quantity of the seal and the water inlet member, thereby reducing the tolerance requirements for each component. Since the tolerance is absorbed, the fluid diverting mechanism will not fail to switch in place. Among them, the water inlet member having elasticity not only has the function of absorbing tolerances, but also acts as a filter screen in the present invention, so as to achieve tolerance absorption and reduce the use of spare parts on the premise of ensuring that the water outlet device has the function of filtering water. Therefore, the product yield is greatly improved under the condition of reducing the production difficulty and production cost.

(76) The above description is only the embodiments of the present invention and do not limit the patent scope of the present invention. Any equivalent transformations made using the content of the description and drawings of the present invention, or the embodiments directly or indirectly applied in related technical fields are also included in the scope of patent protection of the present invention.

Claims

1. A water outlet device, comprising a fluid diverting mechanism, a seal having elasticity, a water inlet mechanism having a water inlet member with elasticity at a water inlet end, and a diversion mechanism rotatably connected to the water inlet mechanism; the water inlet mechanism comprises a water inlet channel; the fluid diverting mechanism comprises a first water outlet channel; the diversion mechanism comprises a second water outlet channel; the seal is mounted to a water inlet end of the fluid diverting mechanism, and the seal is capable of placing the water inlet channel in communication with the first water outlet channel or the second water outlet channel when the diversion mechanism moves relative to the fluid diverting mechanism; when the seal is pressed against a water outlet end of the water inlet mechanism, the water inlet channel communicates with the first water outlet channel; when the seal is pressed against a water outlet side of the water inlet member, the water inlet channel communicates with the second water outlet channel.
2. The water outlet device according to claim 1, wherein the fluid diverting mechanism is installed in the water inlet mechanism and arranged through the axial center of the diversion mechanism; the water inlet mechanism is embedded with an elastic seal at a contact position with the fluid diverting mechanism; when the water inlet channel communicates with the first water outlet channel, the fluid diverting mechanism presses against the elastic seal.
3. The water outlet device according to claim 1, wherein the water inlet member has a water inlet net recessed in an axial direction along a water inlet end of the water inlet mechanism to the water outlet end of the water inlet mechanism; a plurality of water outlet holes are provided around a side wall in the radial direction of the water inlet net.

4. The water outlet device according to claim 3, wherein the side wall in the radial direction of the water inlet net inclines gradually and concavely along the water inlet end of the water inlet mechanism to the water outlet end of the water inlet mechanism.
5. The water outlet device according to claim 1, wherein the fluid diverting mechanism comprises a fluid diverting body; the seal is sleeved outside a water inlet end of the fluid diverting body.
6. The water outlet device according to claim 5, wherein a first limiting stage is provided outside the water inlet end of the fluid diverting body, and the seal is sleeved outside the first limiting stage.
7. The water outlet device according to claim 6, wherein the water inlet end of the fluid diverting body is further provided with a second limiting stage; the first limiting stage and the second limiting stage are provided in sequence along the axial direction of the fluid diverting body; and the seal presses against the second limiting stage.
8. The water outlet device according to claim 5, wherein a side of the seal facing towards a water inlet end of the water inlet mechanism has a first diversion slope; a side of the seal facing towards the water outlet end of the water inlet mechanism has a second diversion slope; the first diversion slope inclines gradually and convexly along the water inlet end of the water inlet mechanism to the water outlet end of the water inlet mechanism; the second diversion slope inclines gradually and convexly along the water outlet end of the water inlet mechanism to the water inlet end of the water inlet mechanism; and when the water inlet channel communicates with the first water outlet channel, the second diversion slope is pressed against the water outlet end of the water inlet mechanism.
9. The water outlet device according to claim 5, wherein the top of the seal and the top of the fluid diverting body form a buffer space in the axial direction of the fluid diverting body.
10. The water outlet device according to claim 9, wherein the buffer space has a height of 0.3 mm to 0.7 mm in the axial direction.
11. The water outlet device according to claim 1, wherein the fluid diverting mechanism comprises a fluid diverting body, a reset member having a first end and a second end, and a functional member; the reset member and the functional member are successively arranged in the fluid diverting body along a water inlet end of the fluid diverting body to a water outlet end of the fluid diverting body; and the first end and the second end of the reset member are pressed against a water inlet end of the water inlet mechanism and the functional member, respectively.
12. A water outlet device comprising a fluid diverting mechanism, wherein the water outlet device further comprises a water inlet mechanism and a diversion mechanism rotatably connected to the water inlet mechanism; a water inlet end of the water inlet mechanism is provided with a water inlet member having elasticity; the water inlet mechanism comprises a water inlet channel; the fluid diverting mechanism comprises a first water outlet channel; the diversion mechanism comprises a second water outlet channel; when the diversion mechanism moves relative to the fluid diverting mechanism, the water inlet channel communicates with the first water outlet channel or the second water outlet channel; when the water inlet channel communicates with the second water outlet channel, the fluid diverting mechanism presses against a side of the water inlet member; wherein the water inlet member has a water inlet net recessed in an axial direction along the water inlet end of the water inlet mechanism to a water outlet end of the water inlet mechanism; a plurality of water outlet holes are provided around a side wall in the radial direction of the water inlet net.
13. The water outlet device according to claim 12, wherein the side wall in the radial direction of the water inlet net inclines gradually and concavely along the water inlet end of the water inlet mechanism to the water outlet end of the water inlet mechanism.
14. The water outlet device according to claim 12, wherein a side of the water inlet net facing towards the fluid diverting mechanism is planar.
15. The water outlet device according to claim 13, wherein a side of the water inlet net facing towards the fluid diverting mechanism is planar.

